# Trust Is Not Agreement

Separating sensor credibility, cross-channel consensus, and physical plausibility in industrial edge fusion


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**Thesis. Sensor trust should be treated as a time-varying evidence state about an observation channel, not as agreement with other sensors. Consensus can strengthen a claim only after independence, operating regime, measurement quality, and physical plausibility are considered.**

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# Abstract

Multisensor condition-monitoring systems often reward agreement among channels, but agreement is not equivalent to sensor health or physical truth. STP-005 proposes a layered Sensor Trust contract that separates intrinsic measurement quality, temporal validity, calibration state, channel history, physical plausibility, cross-sensor support, independence/diversity, and model-specific usefulness. Trust is represented as reason-bearing evidence and converted into claim-specific admission rather than a universal sensor reputation. The architecture is grounded in measurement-uncertainty practice, recent MEMS quality-of-sensing research, sensor/process-fault separation, reliability-aware evidence fusion, and trustworthy virtual sensing. A bounded implementation path is described for SynapticSteel TelePod's Sensor Trust stage. Current TelePod work includes sensor collection, quality/freshness boundaries, and a canonical Sensor Trust stage; the multi-axis trust contract proposed here is not claimed as completed production functionality.


**Evidence boundary. The literature supports explicit measurement uncertainty, sensor-quality assessment, process-versus-sensor fault separation, and reliability-aware fusion. STP-005's exact trust vector, independence guard, and authority rules are an architectural synthesis requiring controlled validation before performance claims.**

# 1. Why agreement is seductive

Suppose six accelerometers are mounted around a machine. Five report similar spectra and one diverges. A consensus-weighted fusion system is tempted to lower the sixth sensor's credibility. Sometimes that is correct: the sensor may be loose, saturated, drifting, or failing. But the same pattern can arise because the sixth sensor is physically closest to an incipient bearing defect. Agreement is evidence about relationships among observations; it is not a direct measurement of truth.

The inverse failure is equally important. Several channels can agree for the wrong reason. Shared power noise, common thermal drift, identical filtering, a network timestamp error, a mechanically coupled mounting artifact, or a common learned preprocessing model can move multiple channels together. Correlated error can look like corroboration.

Modern fusion methods increasingly assign confidence or credibility based on consistency among sources. Dempster-Shafer variants incorporate sensor credibility, and a 2026 Reliability Engineering & System Safety framework splits multisource sensors into confidence groups using cross-sensor fault-information consistency and uncertainty quantification [1]. Such methods can improve fusion, but their success does not make consensus equivalent to sensor health or physical truth.

# 2. Measurement quality precedes model confidence

Metrology provides a useful discipline: a measurement result is incomplete without an account of uncertainty. ISO/IEC Guide 98-1:2024 establishes general rules for evaluating and expressing measurement uncertainty [2]. NIST industrial-wireless testbed work similarly constructs explicit uncertainty budgets for physical and network performance measurements [3]. These traditions treat uncertainty as a property of what is known about a measurement, not as a popularity vote among sensors.

Recent condition-monitoring research is moving in the same direction. Ruhland et al. (2026) describe a continuous Quality of Sensing score for MEMS accelerometers using spectral and time-domain features and physically grounded degradation mechanisms [4]. A 2026 wind-turbine preprint proposes multi-dimensional entropy as a front-end gate for erroneous vibration measurements [5]. Both approaches ask whether the measurement itself is credible before downstream diagnosis.

| Axis | Question | Example evidence |
|---|---|---|
| Intrinsic quality | Is this signal behaving like a valid measurement? | clipping, noise floor, entropy, saturation, spectral artifacts |
| Temporal validity | Is it fresh and correctly time-related? | age, clock state, acquisition window, jitter |

| Axis | Question | Example evidence |
|---|---|---|
| Calibration state | Is scale/orientation/transfer function credible? | calibration provenance, drift checks, mounting/orientation |
| Channel history | Has this channel recently degraded or recovered? | fault/recovery epochs, intermittent dropout, residual history |
| Physical plausibility | Can this value occur under known constraints? | range/rate limits, energy/kinematic/process constraints |
| Cross-sensor support | Do independent observations support it? | coherence, residual agreement, correlated change |
| Independence/diversity | Are supporting witnesses actually distinct? | different modality, power path, preprocessing, mounting |
| Model usefulness | Is this evidence appropriate for this model/task? | operating regime, feature contract, training support |

Table 1. Proposed trust axes. A single scalar can be derived for a bounded task, but the underlying reasons should remain available.

# 3. Trust must be local to a claim

A sensor does not have one universal trust score. A microphone may be excellent evidence for an impulsive acoustic event and weak evidence for shaft displacement. A thermal imager may be temporally slow but physically valuable for sustained heating. An SNMP counter can be highly trustworthy about CPU load while being irrelevant to bearing vibration. Trust is therefore claim-relative.

The proposed architecture stores a reusable evidence vector for the channel, then computes a bounded admission/weight for a particular model or claim. This avoids turning a sensor's historical reputation into permanent authority. A previously degraded sensor can recover; a normally reliable sensor can become inappropriate in a new operating regime.

**Trust vector:** $T_i(t) = \{\text{quality, freshness, calibration, history, plausibility, support, independence}\}$
**Model admission:** $A_{i,m}(t) = g(T_i(t), \text{operating state, model contract, claim})$
The function g may be deterministic, probabilistic, learned, or hybrid. Its inputs and reasons should remain inspectable.

# 4. Disagreement is sometimes the most valuable evidence

Industrial fault-diagnosis literature distinguishes sensor faults from process faults because a false indication and a real process deviation demand different responses. Zhang et al. present an integrated framework separating sensor and process faults [6]. More recent work extends source identification to process faults versus model drift, emphasizing that an anomaly detector must first ask whether the deviation belongs to the process or to the model itself [7].

For physical machinery, localized faults create an analogous ambiguity. One channel can diverge because it is wrong, or because it is the only sensor positioned to see the event strongly. A trust layer should therefore treat disagreement as a trigger for explanation, not automatic demotion. Useful tests include neighboring-channel coherence, operating-state changes, physical propagation delay, mounting geometry, modality diversity, and whether the divergent signal contains internally valid structure.

# 5. Consensus needs an independence guard

Evidence combination is strongest when witnesses are meaningfully independent. In edge systems, apparent sensor multiplicity often hides common dependencies. Two accelerometers may share the same mounting plate. Multiple channels may share a power regulator, USB controller, timestamp source, preprocessing library, or model-derived virtual sensor. A virtual sensor reconstructed from three physical channels is not a fourth independent witness to the same event.

The proposed independence guard attaches dependency lineage to corroboration. Support from channels sharing a known source, transformation, clock, power path, mounting dependency, or derived model should be discounted or explicitly marked as correlated. The goal is not to prove statistical independence; it is to prevent known common causes from masquerading as independent evidence.

# 6. What trust is not

**Historical reputation alone.** Easy to operate, but turns historical performance into permanent authority and adapts poorly to drift, remounting, repair, or regime change.

**Model attention as trust.** Attention can reveal which channels a model uses, but high attention is not proof that a sensor is physically reliable. Recent uncertainty-aware industrial diagnosis research combines process structure and sensor reliability rather than relying on attention alone [8].

**Virtual-sensor substitution.** This can preserve continuity after a physical sensor fault. Recent digital-twin work demonstrates dynamic replacement using learned virtual sensors [9], while physics-residual/conformal approaches seek accuracy, physical consistency, and uncertainty reliability [10]. Substitution should remain explicitly synthetic evidence, not silently inherit the authority of a direct sensor.

**End-to-end learned fusion.** Potentially powerful, but difficult to audit when the training distribution does not include the exact sensor degradation or dependency pattern encountered in service.

# 7. Proposed TelePod Sensor Trust contract

SynapticSteel's canonical Model Flow places Sensor Trust before Operating State and later fusion/reasoning stages. That ordering is technically important: observation credibility should be characterized before the system builds a story from the observations. Existing TelePod work also carries quality/freshness boundaries and explicit missing/stale fallback rules in visualization and sensor-admission design. These are architectural foundations, not evidence that the full STP-005 trust system is implemented.

STP-005 proposes that the Sensor Trust stage emit, for each channel and bounded time window: (1) direct/derived identity; (2) intrinsic quality state and reasons; (3) freshness/timing state; (4) calibration/mounting state; (5) recent reliability history; (6) physical-plausibility result; (7) cross-channel support/conflict; (8) dependency group; and (9) claim-specific admission recommendation. Downstream models should receive the trust evidence alongside features rather than receiving only a pre-weighted scalar.

**Sensor Trust does not diagnose the machine.** It diagnoses the epistemic condition of the observation path sufficiently to govern whether later models may rely on it.

# 8. Controlled evaluation

| Experiment | Question | Expected safe behavior |
|---|---|---|
| Loose one accelerometer | Can trust detect intrinsic/mounting artifact? | Quality/history degrade without declaring machine fault |
| Real localized impact near one sensor | Does disagreement wrongly punish the closest witness? | Divergent channel retained if internally valid and physically plausible |
| Inject shared power/noise artifact | Can consensus be recognized as correlated? | Dependency-aware corroboration prevents false confidence |
| Freeze one stream | Can freshness override plausible scalar values? | Admission degrades as age grows |
| Remount/rotate sensor | Does mounting state invalidate prior trust? | New epoch; prior reputation not silently reused |
| Virtual-sensor substitution | Is synthetic lineage preserved? | Virtual estimate supports continuity but not independent corroboration |
| Operating-regime transition | Does trust confuse regime change with degradation? | Operating-state context limits demotion |

Table 2. Experiments should deliberately separate sensor faults, shared-cause artifacts, and genuine physical disagreement.

Primary metrics should include false sensor-demotion rate during real localized faults, false corroboration rate under common-mode errors, time to detect quality degradation, recovery calibration, confidence calibration, and downstream false-anomaly/false-RCA rates. An ablation should compare scalar consensus trust against the layered trust vector.

# 9. Limitations and counterarguments

**Trust vectors add complexity.** The operator need not see every dimension continuously; the dimensions exist so the system can explain why admission changed.

**Physical plausibility models can be wrong.** They should be bounded constraints, not omniscient simulators. A plausibility failure should reduce authority or trigger investigation, not automatically erase a novel event.

**Independence is difficult to quantify.** Exact statistical independence is unrealistic. The practical goal is dependency awareness: shared hardware, transformations, models, clocks, mounting, and source lineage should be known when possible.

**Consensus remains useful.** STP-005 does not reject agreement; it demotes agreement from truth to one evidence dimension.

**Metrological uncertainty and diagnostic trust are not identical.** GUM-style measurement uncertainty is one foundation, while diagnostic trust additionally includes provenance, freshness, dependencies, operating regime, and task relevance.

# 10. Relationship to prior STP papers

**STP-001 - Failure Metabolism** addresses evidence-preserving operational memory and causal lineage. **STP-002 - Claim Lineage** addresses provenance from physical evidence through transformations, models, and rules to claims. **STP-003 - Temporal Uncertainty Budgets** addresses time-confidence envelopes and limits on temporal correlation. **STP-004 - Silence Is a Signal, But Not a Measurement** addresses missingness as evidence about observability rather than a physical scalar. STP-005 separates agreement from trust.

The common pattern is epistemic restraint: preserve the distinction between what the observation process actually establishes and what a downstream model would like to infer. STP-005 is not a restatement of missingness. A channel can be present, fresh, numerically plausible, and still be untrustworthy because of calibration, mounting, common-mode contamination, or physical inconsistency. Conversely, a channel can disagree strongly with peers and still be the most valuable witness to a localized event.

# 11. Conclusion

Industrial intelligence becomes fragile when "most sensors agree" quietly becomes "the machine must be doing this." Agreement is useful evidence, but its meaning depends on the quality of each measurement, the independence of the witnesses, the operating regime, the physics of the asset, and the claim being made.

A better Sensor Trust layer preserves those distinctions. It scores the observation path before the machine, records reasons rather than only reputation, and lets downstream models know whether support is direct, correlated, synthetic, stale, uncertain, or physically constrained. This does not eliminate uncertainty. It makes uncertainty governable.

**The most trustworthy sensor in the room may be the one disagreeing for the right physical reason.**

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# Related SynapticSteel Technical Papers

**STP-001 - Failure Metabolism.** Evidence-preserving operational memory and causal lineage.
**STP-002 - Claim Lineage.** Provenance from physical evidence through transformations, models, and rules to claims.
**STP-003 - Temporal Uncertainty Budgets.** Time-confidence envelopes and limits on temporal correlation.
**STP-004 - Silence Is a Signal, But Not a Measurement.** Missingness is evidence about observability, not a physical scalar.

# Publication note

STP-005 v1.0 is approved for public release. Zenodo DOI **10.5281/zenodo.23114667** was reserved before finalization and is embedded in this canonical artifact and its metadata. This release supersedes the REVIEW v0.1 artifact. No empirical performance claim is made for the proposed Sensor Trust contract; controlled validation remains future work.